

Cambridge Lower Secondary Checkpoint

MATHEMATICS**0862/01**

Paper 1

October 2025

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

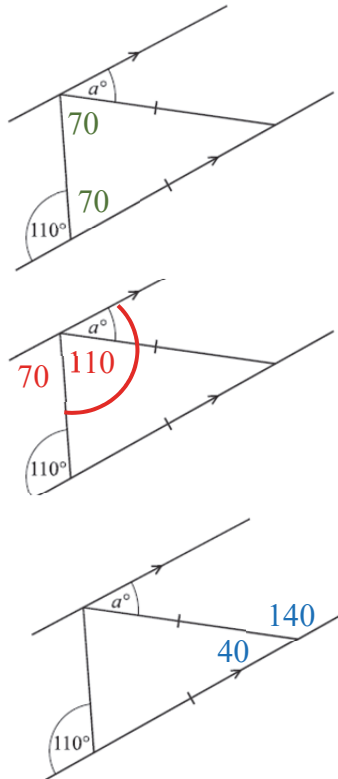
This document has **12** pages.

Mark scheme annotations and abbreviations

FT	follow through after error
SC	special case mark
cao	correct answer only
dep	dependent
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
soi	seen or implied

Question	Answer	Marks	Part marks	Guidance
1	35 800	1		
2	$(-4)^2$ and -4×-2	1		Both answers correct for the mark. Accept any clear indication.
3	vertically opposite	1		Accept reasonable misspellings. Do not accept opposing.
4	An example of relevant quantitative data, e.g. - Amount of money (spent) (Student's) age - The number of students (in each age group).	1		Accept correct equivalents, e.g. - Pricing of food - Scale of 1 to 10 to show how often they buy different foods - The number of lunch orders - How much do you spend? (i.e. presented as a question) - Put the ages in a bar chart (ignore references of what to do with the data, similarly with calculations) - Money (they bring to school). Do not accept qualitative data (also) offered, e.g. Students from different ages - The most popular food - A bar chart of people.
5	84 ($\div 12 =$) 7	2	Award 1 mark for 84 or 7	Correct order only.

Question	Answer	Marks	Part marks	Guidance
6	Correct construction with correct intersecting arcs drawn.	2	Award 1 mark for accurate triangle with no/incorrect arcs or for any equilateral triangle with correct arcs but incorrect size/position or for correct pair of arcs (65 mm) with no/incomplete/unruled triangle.	Accept $\pm 2^\circ$ tolerance. ± 2 mm tolerance. Arcs must be sufficient to complete triangle.
7	$C = 50n + 80$	1		Accept any clear indication.
8	Correct ruled graph of $4x + 5y = 40$ from (0, 8) to (10, 0).	3	Award 2 marks for two correct plots or Award 1 mark for one correct value soi.	Evidence of plotting (4, 5) is max 2 marks. Implied by ends of line drawn. Plots can be other points on the line, e.g. (5, 4). Plots must be indicated if not ends of line. Implied by one correct plot. May be seen in table.

Question	Answer	Marks	Part marks	Guidance
9	$(a =) 40$	3	Award 1 mark for - either of the base angles of the triangle = 70 and Award 1 mark for either of these - alternate angle to 110 = 110 - co-interior angle to 110 = 70 or Award 2 marks for either of these - angle at apex of triangle = 40 - co-interior angle to $a = 140$ or for a full correct method, e.g. $180 - ((180 - 110) \times 2)$ oe $110 - (180 - 110)$ oe.	For part marks: - angles must be marked in the correct place on the diagram or indicated by clear labelling of the diagram - do not apply the choice rule, e.g. one base angle = 70 and the other base angle $\neq 70$ scores 1  1 mark and 1 mark or 2 marks

Question	Answer	Marks	Part marks	Guidance
10	88×1.13	1		Accept any clear indication.
11	Equilateral triangle	1		Accept regular triangle, triangular (base) and reasonable misspellings. Do not accept, e.g. • triangular prism • isosceles triangle.
12(a)	x^9	1		
12(b)	18 and 7	2	Award 1 mark for 18 or 7	Correct order only.
13	= < <	2	Award 1 mark for two correct answers.	
14	16 (seconds)	2	Award 1 mark for 4×20 or for $\frac{4}{5} = \frac{k}{20}$	Or better, e.g. $80, \frac{4}{5} \times 20, 80 \div 5, 20 \div \frac{5}{4}$ Accept a correct full method that involves changing units consistently, e.g. $\frac{4}{3600} \times 20 \times \frac{3600}{5}, \frac{4 \times 10^3}{3600} \times 20 \times \frac{3600}{5 \times 10^3}$ $(4 \div 3.6) \times 20 \div (5 \div 3.6)$

Question	Answer	Marks	Part marks	Guidance
15	-2	2	Award 1 mark for $(-3 + 2)^2 = 1$ oe or better.	Or better, e.g. Award 1 mark for these: 5×1 $5 \times 1 - 7$ Oe, e.g. if brackets expanded first must get to a comparable point such as $(-3)^2 + 2(-3) + 2(-3) + 4 = 1$ or better, e.g. $5(-3)^2 + 20(-3) + 20 = 5$

Question	Answer			Marks	Part marks	Guidance										
16	True ☐ ☒ ☐	False ☒ ☐ ☐	Cannot tell ☐ ☐ ☒	2	Award 1 mark for two correct answers.	Accept any clear indication.										
17	$(x =) 2$ $(y =) 7$			3	Award 1 mark for $(x =) 2$ and Award 2 marks for $(y =) 7$ or Award 1 mark for correct first step, e.g. $8 = 2(y - 3)$ or $\frac{8}{2} = y - 3$	Oe, e.g. $8 = 2y - 6$, $2y = 14$ Award 1 mark for $\frac{8}{7-3} = 2$ (correct solution of 7 shown embedded).										
18	<table><tr><th>True</th><th>False</th></tr><tr><td>☐</td><td>☒</td></tr><tr><td>☒</td><td>☐</td></tr><tr><td>☐</td><td>☒</td></tr><tr><td>☐</td><td>☒</td></tr></table>			True	False	☐	☒	☒	☐	☐	☒	☐	☒	2	Award 1 mark for three correct answers.	Accept any clear indication.
True	False															
☐	☒															
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Question	Answer	Marks	Part marks	Guidance
19(a)	Correct frequency polygon with lines joining the points (20.5, 5), (21.5, 12), (22.5, 17), (23.5, 22) and (24.5, 2).	3	Award 2 marks for a horizontal translation of the correct polygon or Award 1 mark for four or five plots correct horizontally (time = 20.5, 21.5, 22.5, 23.5, 24.5) and Award 1 mark for four or five plots correct vertically (frequency = 5, 12, 17, 22, 2).	Ignore lines to the left of time 20.5 and to the right of time 24.5 Ignore lines to the left of <i>their</i> 20.5 and to the right of <i>their</i> 24.5 Plots may be implied by ends of diagonal lines, otherwise should be a dot or cross. Ignore any bars drawn for full marks or part marks (bars alone or horizontal/vertical lines do not imply plots).
19(b)	4 and 3.1	2	Award 1 mark for two correct and one incorrect or for one correct and none incorrect.	Accept any clear indication.

Question	Answer	Marks	Part marks	Guidance
20(a)	Correct demonstration, e.g. • 10th term is -0.5 • $9 \times 2.5 > 22$ oe • $10 \times 2.5 > 24.5$ oe	2	Award 1 mark for $[9 \times 2.5 =] 22.5$ oe or $22 - 9 \times 2.5$ oe or 19.5, 17, 14.5, 12. 9.5, 7, 4.5, 2 seen or 9 subtractions of 2.5 from 22 soi to a negative number FT one error or $22 - 10 \times 2.5 = -3$ oe or $24.5 - 2.5n$ oe.	For 2 marks, accept repeated subtraction to -0.5 . Subtractions implied by correct terms. Oe, e.g. $\frac{45}{2}$. Accept $-22.5, -\frac{45}{2}$ Oe, e.g. $22 - 22.5, 22 - \frac{45}{2}, \frac{44 - 45}{2}, 22 + 2.5 - 10 \times 2.5, 24.5 - 10 \times 2.5$ Allow equivalent fractions, e.g. $\frac{39}{2}, \frac{34}{2}, \frac{29}{2}, \frac{24}{2}, \frac{19}{2}, \frac{14}{2}, \frac{9}{2}, \frac{4}{2}$ seen. Oe, e.g. $22 - 25 = -3$
20(b)	-8 and -23 and -88	2	Award 1 mark for three correct and one incorrect or for two correct and no incorrect.	Accept any clear indication.
21	102.292 and 42 800	2	Award 1 mark for each correct answer.	Correct order only.

Question	Answer	Marks	Part marks	Guidance
22	2280	3	Award 2 marks for any of these $\left[2 \frac{7}{25} = \right] 2.28$ or $\frac{228}{100}$ $\frac{57}{25} \times 1000$ or better $2 \times 1000 + \frac{7}{25} \times 1000$ or better or Award 1 mark for $\left(9.5 + \frac{1}{2} \right)^3 = 1000$ or $\frac{57}{25}$ oe improper fraction.	Or better, e.g. $\frac{57\,000}{25}$ but not back a step, e.g. $\frac{57}{25} \times 10^3$ scores 1 mark only. Implied by, e.g. $10^3 = 1000$

Question	Answer	Marks	Part marks	Guidance
23	(3, 2)	3	Award 2 marks for x- or y-coordinate correct or Award 1 mark for $(21 - -9) \div 5 \times k$ oe or $(11 - -4) \div 5 \times k$ oe where k is 1, 2 or 3 in either method or for sight of 30 and 12 : 18 or for sight of 15 and 6 : 9	Accept 2 + 3 for 5 and accept equivalents provided they show $(21 - -9)$ or $(-9 - 21)$ or $[-]30$ $(11 - -4)$ or $(-4 - 11)$ or $[-]15$ i.e. oe is not just for 6 or 3 alone or these alone: 6×2, 6×3, 3×2, 3×3, 12, 18, 6 or 9